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Course: Data Structures – Lab 1
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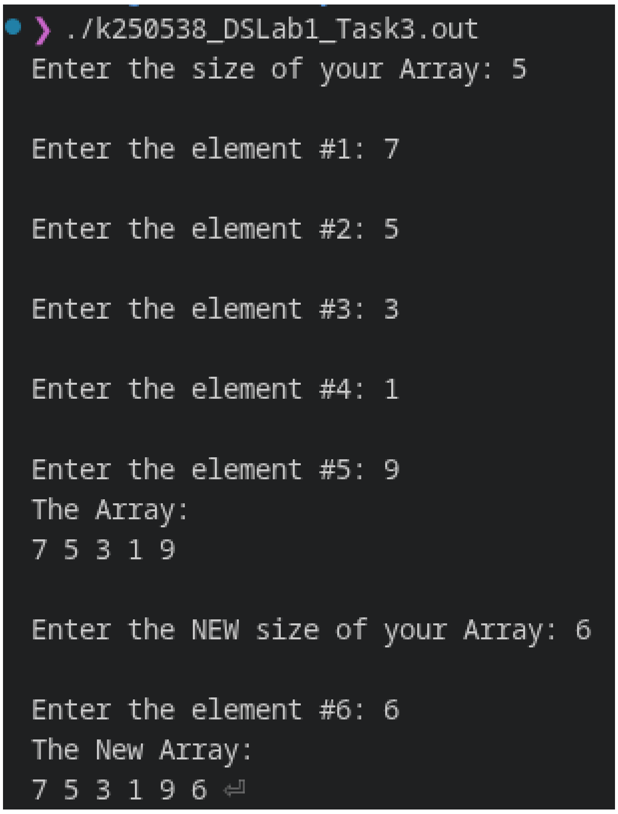
<u>Note: Task 1, 2 and 3 are submitted in the exam submission folder.</u>

Task 4:

```
#include <iostream>
using namespace std;

int main()
{
    cout << "Enter the size of your Array: ";
    int size;
    cin >> size;
    int *array = new int[size];

    for (int i = 0; i < size; i++)
    {
        cout << "\nEnter the element #" << i+1 << ": ";
        cin >> array[i];
    }

    cout << "The Array:" << endl;
    for (int i = 0; i < size; i++)
    {
        cout << array[i] << " ";
    }

    cout << "\n\nEnter the NEW size of your Array: ";
    int new_size;
    cin >> new_size;
    int *new_array = new int[new_size];
    for (int i = 0; i < new_size; i++)
    {
        if (i < size)
        {
            new_array[i] = array[i];
        }
        else
        {
            cout << "\nEnter the element #" << i+1 << ": ";
            cin >> new_array[i];
        }
```

```
    }

    cout << "The New Array:" << endl;
    for (int i = 0; i < new_size; i++)
    {
        cout << new_array[i] << " ";
    }


    delete[] array;
    delete[] new_array;
    return 0;
}
```



Task 5:

```
//25K0538, Muhammad Farhan
#include <iostream>
#include <string>
using namespace std;

class Product
{
   private:
   string name;
   int* quantity;

   public:
   Product(string namae, int quant) : name(namae)
   {
      quantity = new int;
      *quantity = quant;
   }

   Product(const Product &other)
   {
      name = other.name;
      quantity = new int;
      *quantity = *other.quantity;

   }

   ~Product()
   {
      delete quantity;
   }

   Product& operator=(const Product& other)
   {
      if (this == &other)
      {
         return *this;
      }

      name = other.name;

      delete quantity;
      quantity = new int;
      *quantity = *other.quantity;

      return *this;
   }

   void display()
   {
      cout << "Product Name:\t" << name << endl;
      cout << "Quantity:\t" << *quantity << endl;
```

```
    }

};

int main()
{
    Product prod1("soap", 10);
    prod1.display();
    Product prod2 = prod1;
    prod2.display();
    Product prod3("shampoo", 5);
    prod3.display();
    prod3 = prod1;
    prod3.display();

    return 0;
}
```

```
❯ g++ k250538_DSLab1_Task5.cpp -o k250538_DSLab1_Task5.out && ./k250538_DSLab1_Task5.out
Product Name:    soap
Quantity:        10
Product Name:    soap
Quantity:        10
Product Name:    shampoo
Quantity:        5
Product Name:    soap
Quantity:        10
```